

Certification Details



DEEP
LEARNING
INSTITUTE

Rapid Applications Development Using LLMs

The format for demonstrating knowledge in the Instructor Certification interview will be a question and answer session with the expectation that you can demonstrate the ability to field student questions and understand the concepts at a deeper level than those attending a workshop.

The following is not a comprehensive list of topics. You may be asked questions on topics not included below to show your mastery of the subject matter.

To become certified to instruct DLI's **Rapid Applications Development Using LLMs** workshop, you must be prepared to explain the following concepts / terminology:

LlamaIndex	Encoder-only, encoder-decoder, decoder-only	Zero / One / Few shot
LangChain	Runnable in LCEL	LangGraph
Llama / Llama-2 / Llama-3	Tokens / tokenization	NIM
Instruction tuning / alignment	Embeddings	Stochastic Parrot
Auto-regressive forecasting	Agentics (e.g. ReAct) and tools	GPT2, GPT3, ChatGPT, GPT4
Quantization	Whisper Model	T5 model
Transformers and self/cross attention	Query / Key / Value	Masking / MLM
Hugging Face	Foundation Model	BERT
RAG	Hallucinations	CLIP / ViT / VLM / diffusion

Below are some resources that may assist you in your preparations. The resources are not meant to represent an endorsement from NVIDIA nor are they intended to be comprehensive for your studies.

1. [Generative AI – What is it and How Does it Work?](#)
Date unknown
Article
2. [What is Generative AI? Everything You Need to Know](#)
Date unknown
Article
3. [NVIDIA Enables Trustworthy, Safe, and Secure Large Language Model Conversational Systems](#)
Apr 25, 2023
Blog
4. [Implementing Large Language Models](#)
Date unknown
Webcast
5. [Retrieval](#)
Mar 23, 2023
Blog
6. [Getting Started With LlamaIndex](#). Create a Python app to query GitHub.
by Owen Fraser-Green | Better Programming
May 15, 2023
Blog
7. [What is the Difference Between LlamaIndex and LangChain](#)
Sep 10, 2023
Blog
8. [NeMo Guardrails Integration with LangChain](#)
Date unknown
GitHub
9. [Token selection strategies: Top-K, Top-P, and Temperature](#)
May 2, 2023
Blog
10. [Understanding Transformer model architectures - Practical Artificial Intelligence](#)
Feb 13, 2023
Blog
11. [The Ultimate Guide to Deep Learning Model Quantization and Quantization-Aware Training | Deci](#)
Date unknown
Article

12. [Retrieve Anything to Augment Large Language Model](#)
Oct 11, 2023
Research paper
13. <https://huggingface.co/blog/llama2> Llama 2 is here - get it on Hugging Face
Jul 18, 2023
Blog
14. [Understanding Foundation Models: Paving the Way for AI Breakthroughs](#)
Jun 3, 2023
Article
15. [\[2003.07845\] PowerNorm: Rethinking Batch Normalization in Transformers](#)
June 28, 2020
Research Paper
16. [Robust Speech Recognition via Large-Scale Weak Supervision](#)
2022(?)
Research Paper
17. [\[2203.15556\] Training Compute-Optimal Large Language Models](#)
March 29, 2022
Research Paper
18. [Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer](#)
Sep 19, 2023
Research Paper
19. [\[2010.11929\] An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale](#)
June 3, 2021
Research Paper
20. [Power of Large Language Models: The Current State and Future Potential](#)
March 2023
Video
21. [Efficient Inference of Extremely Large Transformer Models](#)
March 2023
Video
22. [Curating Trillion-Token Datasets: Introducing NVIDIA NeMo Data Curator](#)
Aug 8, 2023
Blog
23. [Mastering LLM Techniques: Training](#)
Nov 16, 2023
Blog
24. [Mastering LLM Techniques: LLM Ops](#)
Nov 15, 2023
Blog

25. [Best Practices for Securing LLM-Enabled Applications](#)
Nov 15, 2023
Blog
26. [Retrieval Augmented Generation for Knowledge Intensive NLP Tasks](#)
April 12, 2021
Research Paper
27. [Retrieval Augmented Generation for Large Language Models: A Survey](#)
Dec 18, 2023
Research Paper
28. [What Is Retrieval Augmented Generation aka RAG | NVIDIA Blogs](#)
Nov 15, 2023
Blog
29. [Retrieval Augmented Generation: Streamlining the creation of intelligent natural language processing models](#)
September 28, 2020
Blog
30. [Technical Brief - NVIDIA Docs](#)
Dec 15, 2023
Technical Brief